

Summer Math Adventure

Weeks 6 – 8 Extension Booklet

Exponents · Expressions & Equations · Probability

3 Weeks · 15 Daily Pages · 3 Quizzes · Full Answer Key · ~45 min/day

Name: _____

Summer: 2025

- | | |
|--|--------------------------------------|
| ☐ Week 6: Exponents | ☐ Week 6 Quiz |
| ☐ Week 7: Expressions & Eq. | ☐ Week 7 Quiz |
| ☐ Week 8: Probability | ☐ Week 8 Quiz |

Tools needed: pencil, calculator (Week 6 challenge only)

Summer Math Adventure

W6
D1

Week 6 · Day 1 | What Is an Exponent?

Name: _____

Date: _____

EXPONENTS

Key Idea: An *exponent* tells you how many times to multiply the *base* by itself.

3^4 means $3 \times 3 \times 3 \times 3 = 81$ | base = 3, exponent = 4

Special names: $5^2 =$ "five squared" | $5^3 =$ "five cubed"

Remember: Any number to the power of 0 equals 1. ($7^0 = 1$)

Part A – Label the Parts

For each expression, write the base, the exponent, and read it aloud.

1. 2^5

Base:

Exponent:

Read as:

2. 10^3

Base:

Exponent:

Read as:

3. 6^2

Base:

Exponent:

Read as:

4. 4^0

Base:

Exponent:

Value:

Part B – Expand & Evaluate

Write as repeated multiplication, then find the value.

5. $3^3 = _ \times _ \times _ =$

6. $2^6 = _ \times _ \times _ \times _ \times _ \times _ =$

7. $5^2 = _ \times _ =$

8. $10^4 = _ \times _ \times _ \times _ =$

9. $1^9 =$ _____

10. $4^3 =$ _____ $\times$ _____ $\times$ _____ $=$ _____

Summer Math Adventure

W6
D2

Week 6 · Day 2 · Powers of 10

Part C – Write in Exponential Form

Name: $7 \times 7 \times 7 =$ _____

Date: $12. \underline{9 \times 9} =$ _____

EXPONENTS

Pattern: $10^1=10$ | $10^2=100$ | $10^3=1,000$ | $10^4=10,000$

Each time the exponent goes up by 1, you multiply by 10 (add a zero).

Shortcut: $10^n = 1$ followed by n zeros.

Part A – Complete the Powers of 10 Table

Exponential	Expanded	Standard Form	Number of Zeros
Which is greater: 2^8 or 4^4 ? Show your work and explain.			1
10^2	10×10		
10^3			
10^4			
10^5			
10^6			

Part B – Multiply Using Powers of 10

Use the pattern: $4 \times 10^3 = 4 \times 1,000 = 4,000$

1. $7 \times 10^2 =$ _____

2. $3 \times 10^4 =$ _____

3. $9 \times 10^1 =$ _____

4. $6 \times 10^5 =$ _____

5. $25 \times 10^3 =$ _____

6. $14 \times 10^2 =$ _____

Part C – Fill in the Exponent

7. $10 \boxed{} = 1,000,000$

8. $10 \boxed{} = 100,000$

9. $5 \times 10 \square = 50,000$

10. $8 \times 10 \square = 800$

Summer Math Adventure

Part D – Word Problems

W6
D3

Week 6 · Day 3 | Comparing Exponential Expressions

11. A school has 3×10^2 students. How many students is that?

Name: _____

Date: _____

EXPONENTS

Strategy: Always evaluate each expression first, then compare.

Example: Is 2^5 or 5^2 greater?

$2^5 = 32$ $5^2 = 25$ So $2^5 > 5^2$

Part A – Evaluate Each Side, Then Compare (<, >, =)

1. $3^4 \square 4^3$
 $3^4 = \square$ $4^3 = \square$

★ Challenge

The sun is about 93,000,000 miles from Earth. Write this as a whole number times a power of 10. Show your thinking.

2. $2^7 \square 7^2$
 $2^7 = \square$ $7^2 = \square$

3. $5^3 \square 3^5$
 $5^3 = \square$ $3^5 = \square$

4. $6^2 \square 2^6$
 $6^2 = \square$ $2^6 = \square$

5. $10^3 \square 1,000$

6. $4^2 \square 2^4$

Part B – Order from Least to Greatest

7. 2^3 3^2 2^4 4^1
Values: _____, _____, _____, _____
Order: _____

8. 5^0 2^5 3^3 10^2
Values: _____, _____, _____, _____
Order: _____

Part C – True or False? Correct the False Ones.

9. $2^3 = 6$ T / F
If false, correct: _____

10. $3^0 = 0$ T / F
If false, correct: _____

11. $10^3 = 1,000$ T / F

12. $1^{100} = 100$ T / F
If false, correct: _____

Part D – Pattern Recognition & Apply

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W6
D4

Week 6 · Day 4 · Order of Operations with Exponents

Name: _____

Date: _____

EXPONENTS

PEMDAS Order:

1. Parentheses 2. Exponents 3. Multiplication / Division (left→right) 4. Addition / Subtraction (left→right)

Example: $2 + 3^2 \times 4 = 2 + 9 \times 4 = 2 + 36 = 38$

15. A rumor starts with 1 person. Each day every person tells 3 new people. Write the number of NEW people told on Day 4 as a power.

Day 1: $3^1 = 3$ Day 2: $3^2 = 9$ Day 3: _____
Day 4: _____

_____ new people

14. Which is greater: 2^8 or 8^2 ?

$2^8 =$ _____

$8^2 =$ _____

16. Rank these from least to greatest:

10^4 2^{10} 3^6 4^4

Values: _____, _____, _____, _____

Order: _____

Step 1: _____

Answer: _____

3. 3×2^4

Part E – Cumulative Review (Weeks 1–5)

Step 1: _____
Answer: _____

17. $847 \times 36 =$ _____

5. $2^3 + 3^2$

Step 1: _____ Step 2: _____

Answer: _____

20. $\frac{5}{6} + \frac{3}{4} =$ _____

18. $5,940 \div 45 =$ _____

21. $3.6 \times 4.5 =$ _____

4. $(3 + 1)^2$

Step 1: _____

Answer: _____

6. $50 - 2^2 \times 3$

Step 1: _____ Step 2: _____ Step 3: _____

Answer: _____

19. $\text{GCF}(48, 72) =$ _____

22. Area of rectangle: $l = 9.5$ cm, $w = 6$ cm

$A =$ _____

_____ cm^2

Part B – Multi-Step Expressions

7. $(2 + 3)^2 - 4 \times 3 =$ _____

★ Challenge

Without calculating, explain why 2^{10} must be greater than 10^2 . Hint: think about how fast each grows.

Bonus: Find the value of n if $4^n = 64$.

8. $6^2 \div (3 + 1) + 5 =$ _____

Summer Math Adventure

W6
D5

Week 6 · Day 5 | Perfect Squares & Square Roots

Name: _____

Date: _____

EXPONENTS

Perfect Square: A number that equals an integer times itself. $25 = 5 \times 5 = 5^2$

Square Root ($\sqrt{\quad}$): The opposite — what number times itself gives you this? $\sqrt{25} = 5$

First 12 perfect squares: 1, 4, 9, 16, 25, 36, 49, 64, 81, 100, 121, 144

Part A – Perfect Squares Chart

n	1	2	3	4	5	6	7	8	9	10	11	12
Using the digits 2, 3, and 4 once each (with any operations), write an expression that equals exactly 100. Exponents allowed!												

Part B – Square Roots

1. $\sqrt{36} =$

2. $\sqrt{81} =$

3. $\sqrt{144} =$

4. $\sqrt{49} =$

5. $\sqrt{121} =$

6. $\sqrt{1} =$

Part C – Is It a Perfect Square?

Circle YES or NO. If yes, write its square root.

7. 50 YES / NO $\sqrt{\quad} =$

8. 64 YES / NO $\sqrt{\quad} =$

9. 100 YES / NO $\sqrt{\quad} =$

10. 72 YES / NO $\sqrt{\quad} =$

Part D – Area & Square Roots in Context

11. A square garden has an area of 64 m^2 . What is the length of one side?

Side =

12. A checkerboard has 64 small squares arranged in a perfect square pattern. How many rows does it have?

Rows =

13. If one side of a square tile is 9 cm, what is its area?

Area =

Score: ____ / 13 + chart

Summer Math Adventure

W6
QUIZ

Between which two consecutive whole numbers does $\sqrt{50}$ lie? Explain without a calculator.

Name: _____

Date: _____

Score: _____ / 20 Time started: _____ Finished: _____

Section 1 – Evaluate (1 pt each)

1. $4^3 =$

2. $2^7 =$

3. $9^0 =$

4. $10^5 =$

5. $\sqrt{64} =$

6. $\sqrt{121} =$

Section 2 – Order of Operations (2 pts each)

7. $2^4 + 6 \times 3 =$

8. $(5 - 2)^3 \div 9 =$

9. $5^2 - 3^2 + 4 =$

10. $100 \div 10^2 \times 3 =$

Section 3 – Compare & Reason (2 pts each)

11. Circle the greater value: 3^5 or 5^3 Show your work below.

12. A square room has an area of 196 sq ft. What is the length of each wall? Show work.

Answer:

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W7
D1

Week 7 · Day 1 | Variables & Algebraic Expressions

Name: _____

Date: _____

EXPRESSIONS

Variable: A letter that stands for an unknown number. (x, n, a, b...)

Expression: Numbers, variables, and operations — but NO equals sign. ($3x + 5$)

Term: Parts of an expression separated by + or -. In $3x + 5$: terms are $3x$ and 5

Coefficient: The number multiplied by a variable. In $3x$, the coefficient is 3.

Part A – Vocabulary Match

Label each part of the expression: $4n - 7 + 2n$

Expression: $4n - 7 + 2n$

Number of terms:

List all terms:

Coefficient of the first term:

Constant (number with no variable):

Part B – Write as an Algebraic Expression

Let n = the unknown number.

1. A number plus 8 →

2. 5 times a number →

3. A number divided by 4 →

4. 12 minus a number →

5. Twice a number, plus 3 →

6. A number squared, minus 1 →

Part C – Evaluate the Expression

Substitute the given value and simplify.

7. $3n + 4$ when $n = 5 =$ _____

8. $2x - 7$ when $x = 10 =$ _____

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W7
D2

Week 7 · Day 2 | Simplifying Expressions – Like Terms

Name: _____

Date: _____

9. $m^2 + 1$ when $m = 6 =$ _____

10. $4(a - 3)$ when $a = 8 =$ _____

EXPRESSIONS

Like Terms have the same variable (and exponent). You can combine them by adding/subtracting coefficients.

$3x + 5x = 8x$ | $7a - 2a = 5a$ | **BUT:** $3x + 5y$ cannot be combined — different variables!

Tip: Constants (plain numbers) are also like terms with each other: $4 + 9 = 13$

Part A – Identify Like Terms

Group the like terms by circling them with matching colors (or labels A, B, C...).

1. $3x$ $5y$ $2x$ 8 $4y$ 11 $7x$

Like-term groups:

2. $6a$ $2b$ $3a^2$ $4a$ $9b$ a^2

Like-term groups:

Part B – Combine Like Terms

3. $4x + 3x =$

4. $9y - 2y =$

5. $5a + 2a + 3a =$

6. $8m - 5m + m =$

7. $3x + 4 + 2x + 7 =$

8. $6n - 2 + n + 5 =$

9. $4a + 3b - a + 2b =$

10. $10x + 5 - 4x - 3 =$

Part C – Simplify, Then Evaluate

Week 7 · Day 2 – Simplifying Expressions

Score: _____ / 12

11. $5n + 3 + 2n - 1$ (simplify first, then evaluate when $n = 4$)
Simplified:

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D3

Value:

Week 7 · Day 3 | Writing & Solving One-Step Equations

Name: _____ Date: _____

12. $2x + 2 + x - y$ (simplify first, then evaluate when $x = 3$, $y = 2$)
Simplified:

EQUATIONS

Equation = two expressions set equal. There IS an = sign. $3x + 5 = 20$

Solve = find the value of the variable that makes it true.

Strategy (inverse operations): Whatever is done to the variable, do the opposite to both sides.

$x + 6 = 14 \rightarrow$ subtract 6 from both sides $\rightarrow x = 8$

$3x = 21 \rightarrow$ divide both sides by 3 $\rightarrow x = 7$

★ Challenge

Part A – Name the Inverse Operation

A rectangle has a length of $(3x + 4)$ and a width of $(x + 2)$. Write a simplified expression for the perimeter. Hint: $P = 2l + 2w$.

1. $x + 9 = 15 \rightarrow$ use:

2. $x - 4 = 11 \rightarrow$ use:

3. $5x = 35 \rightarrow$ use:

4. $x \div 3 = 8 \rightarrow$ use:

Simplified perimeter: _____

Part B – Solve. Show Your Check.

5. $x + 12 = 30$

$x =$

Check: $___ + 12 = 30 \checkmark$

6. $y - 7 = 19$

$y =$

Check: $___ - 7 = 19 \checkmark$

7. $4n = 48$

$n =$

Check: $___$

8. $\frac{m}{6} = 9$

$m =$

Check: $___$

9. $t + 25 = 100$

$t =$

10. $8p = 72$

$p =$

Part C – Write the Equation, Then Solve

Score: $_____ / 12$

11. Sara has some stickers. She gives away 14 and has 23 left. How many did she start with?
Equation:

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D4

Answer:

Week 7 · Day 4 | Two-Step Equations

Name: _____

Date: _____

12. A bag of apples costs \$6. Jake buys n bags and spends \$42. How many bags did he buy?
Equation:

EQUATIONS

Two steps: undo addition/subtraction FIRST, then multiplication/division.

Example: $2x + 3 = 11$

Step 1: Subtract 3 from both sides $\rightarrow 2x = 8$

Step 2: Divide both sides by 2 $\rightarrow x = 4$

Always check: $2(4) + 3 = 8 + 3 = 11 \checkmark$

★ Challenge

Part A – Label the Steps

I think of a number, multiply it by 5, then subtract 8. The result is 37. Write and solve the equation. Show all steps.
Solve $5n - 5 = 16$. Fill in the blanks:

Step 1: Add _____ to both sides $\rightarrow 5n = \underline{\hspace{2cm}}$

Step 2: Divide both sides by _____ $\rightarrow n = \underline{\hspace{2cm}}$

Check: $5(\underline{\hspace{1cm}}) - 5 = \underline{\hspace{1cm}} \checkmark$

Part B – Solve. Show All Steps & Check.

1. $2x + 7 = 21$

$x =$

2. $4y - 3 = 17$

$y =$

3. $5m + 10 = 45$

$m =$

4. $\frac{n}{3} + 4 = 10$

$n =$

5. $6a - 12 = 0$

$a =$

6. $3t + 15 = 60$

$t =$

Week 7 · Day 4 · Two-Step Equations

Score: _____ / 8 + Part A

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D5

7. A cab charges a \$5 base fee plus \$3 per mile. The total bill is \$20. How many miles was the ride?

Equation: _____ Date: _____

EQUATIONS

Inequality uses $<$, $>$, $\leq$, $\geq$ instead of $=$. The answer is a range of values.

$x + 3 > 7 \rightarrow x > 4$ (any number greater than 4 works)

Graphing: Open circle $\circ$ for $< / >$ (not equal) · Closed circle $\bullet$ for $\leq / \geq$ (includes the value)

Arrow points in the direction of solutions.

Part A – Write the Inequality

1. "x is greater than 5" $\rightarrow$

2. "y is at most 12" $\rightarrow$

3. "n is less than or equal to 8" $\rightarrow$

4. "t is at least 20" $\rightarrow$

★ **Challenge**
The sum of three consecutive even numbers is 54. If the smallest is n, write and solve an equation to find all three numbers.

Part B – Solve the Inequality

5. $x + 4 > 9 \rightarrow$

6. $y - 3 \leq 7 \rightarrow$

7. $2n < 16 \rightarrow$

8. $\frac{m}{4} \geq 3 \rightarrow$

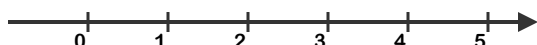
9. $3x - 1 > 11 \rightarrow$

10. $5 + 2y \leq 19 \rightarrow$

Part C – Graph the Solution on the Number Line

Draw an open or closed circle, then shade the arrow.

11. $x > 3$



12. $y \leq 2$



Score: _____ / 12

★ Challenge

Summer Math Adventure

W7
QUIZ

You have at most \$25 to spend. Snacks cost \$4 each. Write and solve an inequality for the maximum number of snacks you can buy. Graph the solution.

Week 7 Quiz | Expressions & Equations

Name: _____ Date: _____

Score: _____ / 20 Time started: _____ Finished: _____

Section 1 – Expressions (1 pt each)

1. Write an expression: "6 less than 3 times x"

2. Evaluate $4n - 2$ when $n = 7$ =

3. Simplify: $5x + 3 + 2x - 1 =$

4. Simplify: $6a - 2b + a + 3b =$

Section 2 – Equations (2 pts each)

5. $x + 17 = 40$

x =

6. $7y = 63$

y =

7. $3m + 8 = 29$

m =

8. $\frac{n}{4} - 2 = 5$

n =

Section 3 – Word Problem & Inequality (2 pts each)

9. A box of crayons costs \$5. Lena also buys a notebook for \$3. She spends \$23 total. How many boxes of crayons did she buy?

Equation:

Name: _____ **Date:** _____

Probability = how likely an event is to happen.

Scale: 0 (impossible) ←————→ 1 (certain)

Place each event on the probability scale below. Write the letter at the right spot.



Percent: _____

Percent: _____

Summer Math Adventure

W8
D2

Week 8 · Day 2 | Experimental vs. Theoretical Probability %

Name: _____

Date: _____

PROBABILITY Write Your Own Probability Scenario

Theoretical Probability: What SHOULD happen based on math. $P(\text{heads}) = \frac{1}{2}$

Experimental Probability: What ACTUALLY happens when you do the experiment.

If you flip a coin 20 times and get 12 heads $\rightarrow P(\text{heads}) = \frac{12}{20} = \frac{3}{5}$

The more trials you run, the closer experimental gets to theoretical.

Part A – Coin Flip Experiment Data

Omar flipped a coin 30 times. Results are below.

Outcome	Tally	Frequency
Heads		1
Tails		

10. A standard deck of 52 cards has 4 suits (hearts ♥, diamonds ♦, clubs ♣, spades ♠), each with 13 cards.

a) $P(\text{heart}) =$ _____

3. Theoretical $P(\text{Heads}) =$ _____

4. Are they the same? Why/why not? _____

c) $P(\text{red card}) =$ _____

Part B – Die Roll Experiment

A die was rolled 60 times. Fill in the theoretical, then compare.

Number	1	2	3	4	5	6
Experimental Freq.	12	8	11	9	13	7
Experimental Prob.						
Theoretical Prob.						

5. Which number had experimental probability closest to theoretical?

Part E – Complement Rule

~~$P(\text{event does NOT happen}) = 1 - P(\text{event})$. This is called the complement.~~

11. $P(\text{rolling a 4 on a die}) = \frac{1}{6}$

12. $P(\text{picking a vowel from MATHEMATICS}) =$ _____
6. If you rolled 600 times instead of 60, would you expect experimental to be closer to or farther from theoretical? Explain. _____

Part C – Compare and Analyze

7. A bag has 5 red and 5 blue marbles. You draw 20 times (replacing each time). Results: 13 red, 7 blue. $P(\text{rain today}) = 0.35$, then $P(\text{rain today}) =$ _____

14. A bag has only red and blue marbles

$$P(\text{red}) = \frac{3}{7}$$

$$P(\text{blue}) =$$

W8
D3

Week 8 · Day 3 | Sample Space & Tree Diagrams

Experimental $P(\text{red}) =$ _____

Date: _____

PROBABILITY

Sample Space: All possible outcomes of an experiment.

Tree Diagram: A visual tool to list all outcomes systematically.

Example – Flip a coin AND roll a die:

H → 1, 2, 3, 4, 5, 6 T → 1, 2, 3, 4, 5, 6 Total: 12 outcomes

Counting Principle: # of outcomes = options × options × ...

Part A – List the Sample Space

1. Flip a coin (H/T) and spin a spinner with colors Red, Blue, Green.

Complete the sample space:

{ _____ }

Total outcomes: _____

8. Marcus spins a spinner 50 times. The spinner has 4 equal sections: A, B, C, D.

He gets: A = 15, B = 11, C = 14, D = 10.

Part B – Draw a Tree Diagram

2. A lunch special has 2 sandwich choices (Turkey, Veggie) and 3 drink choices (Water, Juice, Milk). Draw a tree diagram and list ALL combinations.

$P(A)$ experimental = _____

b) Which result is closest to theoretical?

Total combinations: _____

c) Which result is farthest from theoretical?

Part C – Use the Counting Principle

3. A shirt comes in 4 colors and 3 sizes. How many combinations?

4. You roll 2 dice. How many total outcomes?

Part D – Cumulative Review Sprint

~~Keep your skills sharp! Solve these from earlier weeks.~~

9. Evaluate: $3^4 - 2^5 =$ _____

10. Solve: $4x + 7 = 31$

11. $\frac{7}{8} \times \frac{4}{5} =$ _____

5. A password uses 1 letter (A–E) and 1 digit (1–5). How many passwords?

6. You flip 3 coins. How many total outcomes?

12. Volume of box: $8 \times 3 \times 5 =$ _____

13. What quadrant is $(-3, 4)$ in?

14. $0.6 \times 0.07 =$ _____

Part D – Probability from a Sample Space

Use the Turkey/Veggie × Water/Juice/Milk sample space from Part B.

7. $P(\text{Turkey AND Juice}) =$ _____

8. $P(\text{any Veggie combo}) =$ _____

9. P(any Milk combo) = _____

10. P(Veggie AND Water) = _____

★ **Challenge**

Summer Math Adventure

Design your own experiment to describe it, state the theoretical probability of your chosen outcome, run it 20 times (or simulate with a pencil), and record results. Did your results

W8
D4

Week 8 · Day 4 | Making Predictions with Probability

★ **Challenge**

Name: _____ Date: _____

PROBABILITY

A game show has 3 doors. Behind one is a prize. You pick Door 1. The host opens Door 3 to show no prize. Should you switch to Door 2? Use probability to explain your reasoning.

Prediction Formula: Expected outcomes = $P(\text{event}) \times \text{number of trials}$

Example: $P(\text{rolling a 3}) = \frac{1}{6}$. If you roll 60 times: expected 3s = $\frac{1}{6} \times 60 = 10$

This is a *prediction*, not a guarantee — actual results may vary!

Part A – Calculate Expected Outcomes

A spinner has 5 equal sections: 1 Red, 1 Blue, 2 Yellow, 1 Green.

1. $P(\text{Red}) =$

2. $P(\text{Yellow}) =$

3. If you spin 50 times, expected # of Red:

4. If you spin 50 times, expected # of Yellow:

5. If you spin 100 times, expected # of Green:

6. If you spin 30 times, expected # of Blue:

Part B – Real-World Predictions

7. A factory knows 3 out of every 200 light bulbs are defective. If they produce 1,000 bulbs, how many do they expect to be defective?

Answer:

8. A survey of 50 students showed 20 prefer math class. If there are 350 students in the school, how many would you predict prefer math?

Answer:

9. A basketball player makes 3 out of every 4 free throws. If she attempts 64 free throws in a season, how many does she expect to make?

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W8
D5

Week 8 · Day 5 | Probability Review & Mixed Practice

Name: _____

Date: _____

PROBABILITY

Week 8 Summary:

- ✓ $P(\text{event}) = \text{favorable} \div \text{total}$ | ✓ $0 = \text{impossible}$, $1 = \text{certain}$
- ✓ Theoretical vs. Experimental | ✓ Tree diagrams = list all outcomes
- ✓ Predictions = $P \times \text{trials}$ | ✓ Complement: $P(\text{NOT event}) = 1 - P(\text{event})$

Part A – Mixed Probability Problems

Answer:

A standard deck has 52 cards: 4 suits ($\spadesuit$ $\heartsuit$ $\diamondsuit$ $\clubsuit$), 13 cards each, including Ace, 2–10, Jack, Queen, King.

1. $P(\text{drawing a heart}) =$ _____

2. $P(\text{drawing a King}) =$ _____

★ Challenge

3. $P(\text{drawing a red card}) =$ _____
A bag has some red and blue marbles. You draw 40 times (with replacement) and get red 15 times. Estimate how many red marbles are in the bag if there are 8 total. Show your reasoning. _____

4. $P(\text{NOT a spade}) =$ _____

5. $P(\text{a Jack or Queen}) =$ _____

6. $P(\text{a number card 2–9 of clubs}) =$ _____

Part B – Complement Rule

$P(\text{NOT event}) = 1 - P(\text{event})$

7. $P(\text{rain}) = 0.3$. $P(\text{no rain}) =$ _____

8. $P(\text{win}) = \frac{2}{5}$. $P(\text{not win}) =$ _____

9. $P(\text{red marble}) = \frac{3}{8}$. $P(\text{not red}) =$ _____

10. $P(\text{NOT rolling a 6}) =$ _____

Part C – Scenario & Decision

11. A raffle sells 200 tickets. You buy 5. What is the probability you win?

$P(\text{win}) =$ _____

12. Is a probability of $\frac{7}{10}$ closer to impossible or certain? Place it on the scale and explain.

Summer Math Adventure

W8
QUIZ

Week 8 Quiz | Probability

Name: _____

Date: _____

Score: _____ / 20 Time started: _____ Finished: _____

★ Challenge – Connect All Three Topics

Section 1 – Probability Calculations (1 pt each)

A box has cards numbered 1–10. You draw one card. (a) What is P(drawing a perfect square)? (b) If you draw 50 times (with replacement), how many perfect squares do you expect? (c) Write an expression for expected perfect squares if you draw x times.

4. P(NOT red) = _____

5. P(red or blue) = _____

6. P(green) = _____

Section 2 – Tree Diagram & Counting (2 pts each)

7. You choose a flavor (Chocolate, Vanilla) and a cone (Waffle, Sugar, Cup). Draw a tree diagram and list all combinations.

Total: _____

8. From that same sample space, what is P(Chocolate AND Waffle cone)?
P = _____

Section 3 – Prediction & Reasoning (2 pts each)

9. P(landing on blue) = $\frac{1}{4}$. If you spin 80 times, how many blues do you expect?

Answer: _____

10. A survey found 35 out of 100 students walk to school. Predict how many of 400 students

Week 8 Quiz – Probability

Total: _____ / 20

Answer Key – Weeks 6, 7 & 8

For parent/teacher use only. Cut or fold before giving booklet to student.

WEEK 6 – EXPONENTS

Day 1 – What Is an Exponent?

Part A:

- base=2, exp=5, "two to the fifth power"
- base=10, exp=3, "ten cubed"
- base=6, exp=2, "six squared"
- base=4, exp=0, value=1

Part B:

- $3 \times 3 \times 3 = 27$
- 64
- 25
- 10,000
- 1
- 64

Part C:

- 7^3
- 9^2
- 2^5
- 10^3

Challenge: $2^8=256$, $4^4=256$ — equal!

Day 2 – Powers of 10

Table: 100; $10 \times 10 \times 10 = 1,000$; 10,000; 100,000; 1,000,000

Part B:

- 700
- 30,000
- 90
- 600,000
- 25,000
- 1,400

Part C: 7. 6 8. 5 9. 4 10. 2

Part D: 11. 300 12. 200,000 13. 300,000,000

Day 3 – Comparing Expressions

Part A:

- $81 > 64$ so $>$
- $128 > 49$ so $>$
- $125 < 243$ so $<$
- $36 < 64$ so $<$
- $=$
- $=$

Part B:

- 8, 9, 16, 4 $\rightarrow$ 4, 8, 9, 16
- 1, 32, 27, 100 $\rightarrow$ 1, 27, 32, 100

Part C:

- F (=8)
- F (=1)
- T
- F (=1)

Day 4 – PEMDAS with Exponents

Part A:

- $5+8=13$
- $16-10=6$
- $3 \times 16=48$
- $(4)^2=16$
- $8+9=17$
- $50-12=38$

Part B:

- $25-12=13$
- $36 \div 4 + 5 = 14$
- $27-18=9$

Part C:

- $(3+5)^2 \times 2 = 128$
- $4 \times (2+1)^2 = 36 \neq 9$; try $4 \times 2 + 1 = 9$ ✓

Day 5 – Perfect Squares & Square Roots

Chart:

1, 4, 9, 16, 25, 36, 49, 64, 81, 100, 121, 144

Part B: 1.6 2.9 3.12 4.7 5.11 6.1

Part C: 7.No 8.Yes, 8 9.Yes, 10 10.No

Part D: 11. 8m 12. 8 rows 13. 81 cm^2

Challenge: $7^2=49$, $8^2=64$, so $7 < \sqrt{50} < 8$

Week 6 Quiz

Sec 1: 1.64 2.128 3.1 4.100,000 5.8 6.11

Sec 2: 7. $16+18=34$ 8. $27 \div 9=3$ 9. $25-9+4=20$ 10. $100 \div 100 \times 3=3$

Sec 3: 11. $3^5=243 > 5^3=125$; circle 3^5 12. $\sqrt{196}=14$; each wall = 14 ft

WEEK 7 – EXPRESSIONS & EQUATIONS

Day 1 – Variables & Expressions

Vocab: 3 terms; $4n$, $2n$, -7 ; coeff=4; const= -7

Part B: 1. $n+8$ 2. $5n$ 3. $n \div 4$ 4. $12-n$ 5. $2n+3$ 6. n^2-1

Part C: 7.19 8.13 9.37 10.20 11.12 12.16

Challenge: $3+2h$; 4 hrs = \$11

Day 2 – Like Terms

Part B: $3.7x$ $4.7y$ $5.10a$ $6.4m$ $7.5x+11$
 $8.7n+3$ $9.3a+5b$ $10.6x+2$

Day 3 – One-Step Equations

Part A: 1.subtract 9 2.add 4 3.divide by 5
4.multiply by 3

Day 4 – Two-Step Equations

Label: add 5; $3n=21$; $\div 3$; $n=7$; $3(7)-5=16$ ✓

Day 5 – Inequalities

Part A: $1.x>5$ $2.y\leq 12$ $3.n\leq 8$ $4.t\geq 20$

Week 7 Quiz

Sec 1: $1.3x-6$ 2.26 $3.7x+2$ $4.7a+b$

Sec 2: $5.x=23$ $6.y=9$ $7.m=7$ $8.n=28$

Sec 3: $9.5c+3=23$, $c=4$ boxes $10.x>4$; e.g. $x=5$ works because $4(5)-6=14>10$ ✓

WEEK 8 – PROBABILITY

Day 1 – Introduction

Scale: $A=0$, $B=\frac{1}{2}$, $C=1$, $D=\frac{1}{2}$, $E=0$, $F=\frac{1}{2}$

Day 2 – Experimental vs. Theoretical

Coin: Heads=16, Tails=14; Exp
 $P(H)=16/30=8/15$; Theo $P(H)=1/2$; not same
due to random variation

Day 3 – Sample Space & Tree Diagrams

Part A: {H-R, H-B, H-G, T-R, T-B, T-G}; 6
outcomes

Part B: 6 combinations (2×3)

Day 4 – Predictions

Part A: $1.1/5$ $2.2/5$ 3.10 4.20 5.20 6.6

Day 5 – Review

Part A (deck): $1.13/52=\frac{1}{4}$ $2.4/52=1/13$
 $3.26/52=\frac{1}{2}$ $4.39/52=\frac{3}{4}$ $5.8/52=2/13$
 $6.8/52=2/13$

Part C: $11.7n+2$; when $n=4$: 30 $12.4x+y$;
when $x=3, y=2$: 14

Challenge: $P=2(3x+4)+2(x+2)=8x+12$

Part B: $5.x=18$ $6.y=26$ $7.n=12$ $8.m=54$ $9.t=75$
 $10.p=9$

Part C: $11.s-14=23$, $s=37$ $12.6n=42$, $n=7$

Challenge: $5n-8=37$, $n=9$

Part B: $1.x=7$ $2.y=5$ $3.m=7$ $4.n=18$ $5.a=2$
 $6.t=15$

Part C: $7.3m+5=20$, $m=5$ miles $8.8w+12=60$,
 $w=6$ weeks

Challenge: $n+(n+2)+(n+4)=54$; $n=16$; nums:
16, 18, 20

Part B: $5.x>5$ $6.y\leq 10$ $7.n<8$ $8.m\geq 12$ $9.x>4$
 $10.y\leq 7$

Challenge: $4s\leq 25$, $s\leq 6.25$, so max 6 snacks

Part B: $1.3/12=\frac{1}{4}$ $2.4/12=\frac{1}{3}$ $3.5/12$ $4.9/12=\frac{3}{4}$
 $5.7/12$ 6.0

Challenge: $P(\text{red})=3/8$; $40\times 3/8=15$ times
expected

Die table: Each theoretical = $1/6$ (or $10/60$);
experimental varies; closer with more trials

Part C: 3.12 4.36 5.25 6.8

Part D: $7.1/6$ $8.3/6=\frac{1}{2}$ $9.2/6=\frac{1}{3}$ $10.1/6$

Part B: 7.15 bulbs 8.140 students 9.48 free
throws 10.60 rolls

Part B: 7.0.7 8.3/5 9.5/8 10.5/6

Part C: 11. $P(\text{win})=5/200=1/40$; $P(\text{lose})=39/40$
12. closer to certain

Challenge: $P(\text{perf sq})=3/10$ (1,4,9);
expected=15; expression= $3n/10$

Week 8 Quiz

Sec 1: $1.6/10=3/5$ $2.3/10$ $3.1/10$ $4.4/10=2/5$ $5.9/10$ 6.0

Sec 2: 7. 6 combinations (Choc-Waffle, Choc-Sugar, Choc-Cup, Van-Waffle, Van-Sugar, Van-Cup)

8. $P(\text{Choc AND Waffle})=1/6$

Sec 3: 9. $\frac{1}{4} \times 80 = 20$ blues 10. $35/100 \times 400 = 140$ students